

Allometry of Major CNS Divisions: Towards a Reevaluation of Somatic Brain-Body Scaling

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Abstract. For each of four rodent species, average weights for the brainstem, cerebellum, and forebrain were determined and average cross-sectional area of the cervical spinal cord (SCA) was calculated. Linear regression analyses against log body weight were performed on these data (log translated), along with data (except SCA) from the literature for insectivores and primates. Results indicate that the different CNS divisions scale by different powers of body weight in different mammals and that the rodent SCA varies by less than the 2/3 power of body weight. Based on the results, we conclude that (1) there are at least two general brain scaling factors, somatic and nonsomatic, that necessitate a more complex general allometric equation; (2) the exponent for somatic brain scaling is approximately 0.52, and (3) body surface area is not a primary determinant of brain size.

Introduction

Traditionally, brain allometry research has centered around fitting brain/body weight data to the equation:

$$y = bx^a \text{ [as proposed by Snell, 1891], or} \quad (1)$$

$$E = kP^a \text{ [the version used by} \quad (2)$$

$$\text{Jerison, 1973], or} \quad (3)$$

$$\log(E) = \log(k) + a \log(P) \text{ [the log translation]}$$

where $y = E$ = brain weight, $x = P$ = body weight, a = exponent of allometry, and

$b = k$ = coefficient of cephalization. According to Snell [1891], the exponent, a , represents the relationship between body weight and that part of a brain's weight that may be attributed to 'somatic' factors. The coefficient, b , termed the 'psychic factor', reflects the contribution to brain weight from nonsomatic sources, such that for any animal the higher the value of b , the more 'cephalized' the animal is. Usually, the log form of the equation (3) is used for ease of statistical handling. The

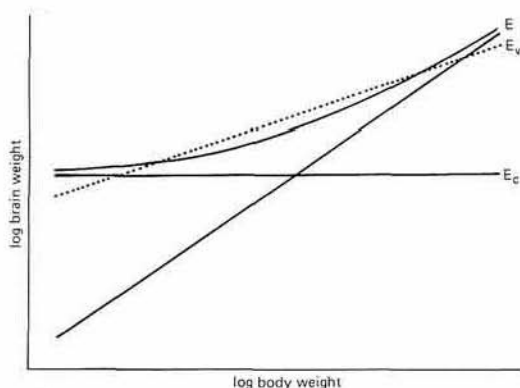


Fig. 1. A hypothetical set of curves for E_c , E_v , and their sum, E , all plotted against body weight, P , on a log/log scale. ... = regression line for the log E plot.

value of a is then represented by the slope of the curve and b (or k) by its y-intercept. For individual animals, k (termed the 'coefficient of cephalization') refers to the deviation from the brain weight-body weight curve along the log brain weight dimension. Specifically, k represents the actual brain weight, as a proportion of the expected brain weight, based on somatic scaling.

More recently, Jerison [1973] [also see Jerison, 1985; Hofman, 1982; Lashley, 1949; and Portmann and Stingelin, 1961] has also considered possible somatic and nonsomatic contributions to brain allometry. According to Jerison [1973], the total brain weight, E , may be considered the sum of 'two independent components, E_v , determined by body size, and E_c , associated with improved adaptive capacities' (p. 78; *italics added*):

$$E = E_c + E_v. \quad (4)$$

Here, E_v is the sum total of the weights of

all neurons (and associated glia) or areas necessary 'to maintain vegetative, sensorimotor, and related behavior' (p. 79). E_c is the sum total of the weights of all 'additional neurons' (p. 79) or areas not essential to maintenance of somatic functioning. Hereafter, in order to simplify terminology we shall refer to these two classes of neurons and/or brain areas as being somatic and nonsomatic respectively.

Jerison (p. 79) defends that E_c may indirectly affect E_v , because an increase in the number of 'additional neurons', N_c , would result in an increase in E , which would be associated with a decrease in neuron density. Thus for the number of somatic neurons, N_v , to remain constant, E_v would have to be greater. This is directly contradictory to his statement (p. 78) that E_c and E_v are independent and to his notion that it is the volume, not the number of neurons, that determines the 'information processing capacity' of brain tissue. Similarly, any effect of E_v on E_c would be contrary to Jerison's theories. Since the effect of E_c on E_v is a minor point, we shall assume that Jerison holds that E_c and E_v are mutually independent and, furthermore, that E_c is independent of body weight, P , of which E_v is a function.

If E_c is independent of body weight, the plot of log E_c , as a function of log P should have the slope, $a_c = 0$ (see fig. 1). On the other hand, E_v should vary with P according to equation 3 (substituting E_v for E), yielding a log/log regression line with a slope $a_v > 0$. But the sum, E , of E_v and E_c does not yield a simple exponential relationship; accordingly, the predicted log/log plot of E , as a function of P , is not linear. Instead, it approaches the plot of log

E_v asymptotically for $E_v > E_c$, and it approaches that of $\log E_c$ for $E_v < E_c$. As a result, a regression line along any segment of the function, has a slope, a , such that $a_c < a < a_v$. It should be noted that a must always range between a_c and a_v , even if a_c is actually greater than a_v .

If there are any discrepancies between a , a_c , and a_v , then traditional approaches to describing brain allometry may be problematic. One major problem may occur as a result of the possible curvature in the $\log E$ function. Traditionally, linear regression lines have been fitted to this function, in order to describe somatic variation and to estimate k for different species. But if the $\log E$ function is actually nonlinear, then use of a linear regression line necessarily introduces systematic error into the estimation of $\log E$. Specifically, the regression line underestimates $\log E$ for very large and very small animals and overestimates $\log E$ for moderately sized animals (see fig. 1). Consequently, estimates of k and of E_c (which depend on accurate estimations of E) are systematically affected. The above problems may be resolved by using the *true* $\log E$ function (whatever it may be) in order to predict $\log E$ and to estimate E_c and k . But to use the $\log E$ function properly requires an understanding of how E_c varies with P , because systematic variation in E_c results in systematic variation in $\log E$ from *nonsomatic* sources, which systematically affects estimates of E_c and k .

The use of $\log E_v$, as a function of $\log P$, may be a less problematic approach to estimating E_c and k , as it is more likely to be a linear function and, therefore, may be approximated better with a linear regression line. Furthermore this function does

not represent systematic variation in either E_c (equal to 0) or k (equal to 1), despite the values of a_c and a_v . Therefore, there should be no systematic error in estimating E_c and k for individual species. Of course, the $\log E_v$ function is more difficult to derive than the $\log E$ function, as somatic and nonsomatic factors must be distinguished.

In order to assess whether there is any curvature in the $\log E$ function, and in order to determine exactly how various allometry functions may be utilized for the estimation of E_c and k , it is necessary to understand the relationships between a , a_c , and a_v . Without this understanding, estimates of k and E_c have little meaning, except where animals of similar body weights are compared. In order to examine these exponents, it is first necessary to attempt experimentally to isolate E_c from E_v . If it is found that the allometry exponents do not differ from each other, then the $\log E$ regression line may be used validly to predict $\log E$ and possibly to estimate k and E_c . Otherwise, the allometric equations and mathematical methodology must be reexamined and revised.

By examining and comparing the allometry of different divisions of the CNS [see Armstrong, 1985; Bennett and Harvey, 1985; Hofman 1982; Holloway, 1979; Sacher, 1970; Wimer, 1979], it may be possible to differentiate the effects of a_v and a_c on the allometry of the whole brain. Undoubtedly, it is difficult or even impossible to divide the brain into regions that are completely somatic or completely nonsomatic [see Jerison, 1985]. However, it should be very easy to divide the brain into regions that *differ* in their somatic and nonsomatic content. Presumably, if a_v differs from a_c , then these regions should dif-

fer in their exponents of allometry, such that regions that are mostly somatically oriented would scale by exponents closer to a_v , and regions that are mostly nonsomatically oriented would scale by exponents closer to a_c . Furthermore, if all regions scale by the same exponent [as Jerison, 1970, 1985, predicts], then we may conclude that a_v and a_c are equal to that exponent. In our study, allometric relationships between body weight and whole brain weight, forebrain weight, cerebellum weight, and brainstem weight were determined in rodents, basal insectivores, progressive insectivores, prosimians, and simians. Nonrodent data were obtained from Stephan et al. [1970].

The secondary goal of this study is to reevaluate the hypothesis that brain size scales in direct proportion to the surface area of the body [Jerison, 1973, 1979]. This theory has previously been questioned [see Armstrong, 1982, 1983, 1985; Martin, 1981, 1982; Szarski, 1980] for two reasons. The first reason is that as more and more mammalian species are examined, the exponent of allometry for the whole brain seems to be approximately 0.75 [Hofman, 1982; Martin, 1981, 1982], rather than 0.67 (the exponent by which body surface area varies with body weight). But if surface area did, in fact, determine somatic brain size variation, then it is a_v that should equal 0.67, not necessarily a_c , which could equal almost any value, depending on nonsomatic brain/body relationship. The second reason that the surface area hypothesis has been questioned is that a few studies tentatively indicate that the density of body surface innervation does not seem to be constant, as body size varies. For instance, Straile [1969] indicates that the

density of sensory epithelial pads differs among four species, including man. A different approach has been to compare the numbers of fibers contained in various nerves and tracts to body weight. These studies [e.g. Bruesch and Arey, 1942; Lassek, 1942; and Towe, 1973] have not demonstrated scaling to the $2/3$ power of surface area. But because these types of studies are scarce and problematic, further work is needed.

To investigate the possibility of surface area scaling, we examined the cross-sectional area of the spinal cord (SCA) for four rodent species, as a convenient indicant of the total number of somatic afferents and efferents. Hofman [1982] indicates that the SCA accounts for 99.2% of the variance in body weight, among 14 species taken from Stephan et al. [1970]. Because this measure accounted for the highest degree of variance of all his measures, Hofman concluded it was a good indicant of body size. Likewise, we feel that it should represent somatic scaling better than any single fiber tract cross-sectional area (although it is not known whether it represents an improvement over fiber counts, except in convenience). Rodents are used in this study partially because Jerison [1979] predicts that this order should exhibit $2/3$ exponential scaling, as a consequence of certain tentatively determined characteristics of cortical organization within the order. Our prediction is as follows: if total somatic innervation is directly proportional to body surface area, then it should be directly proportional to the $2/3$ power of body weight; thus, if the SCA is directly proportional to the total fiber count, then it should also scale by the $2/3$ power of body weight.

Materials and Methods

Measurements were performed on the following animals: 22 male and 22 female CD-1 mice, 8 male Mongolian gerbils (*Meriones unguiculatus*), 17 male hamsters (*Cricetus cricetus*), and 8 female Sprague-Dawley rats. All animals were mature and healthy, and were killed either by asphyxiation in CO₂ or by an overdose of pentobarbital. Immediately subsequent to death, animals were weighed, using a 1/10 g animal scale, and decapitated as far caudally as possible. Heads were then placed in 10% formalin for approximately 3–5 days. (Fixation of such small heads was quite satisfactory without perfusion.) Because this study is only concerned with allometric exponents, fixation in formalin does not adversely affect the results, as the power functions are altered not in shape, but in displacement along the log brain weight dimension only. After fixation, brains and spinal cords were removed (taking care to preserve the integrity of the olfactory bulbs), and the meninges were peeled away. The spinal cords were then cut from the brains at their rostral-most extent. With a cryostat, 20- μ m sections were taken at the spinal cords' rostral ends, and the sections were wet-mounted on slides. Tracings were made of the projected images (40 $\times$) of the slides, and the tracings were weighed using a single-pan analytical balance. The weights were then translated into the corresponding area measurements (SCA).

The whole brains (WB) were then weighed using an analytical balance. Then each cerebellum (CB) was removed and weighed. The remainder of each brain was cut midsagittally, and the halves were cut between the mesencephalon and diencephalon. Specifically, cuts were made to a depth extending just to the neocortex, perpendicular to the midsagittal plane, and extending from the dorsal depression immediately rostral to the colliculi to the ventral depression immediately caudal to the mammillary bodies. The total forebrain (FB) and total brainstem (BS) weights were obtained by weighing the complementary halves together. Data were not taken on any damaged brain parts.

Data for 22 insectivore, 20 prosimian, and 21 simian species were obtained from Stephan et al. [1970]. Stephan et al. categorized the insectivores in two groups, 11 basal insectivores and 11 progressive insectivores. We preserve this distinction for data analysis. Stephan et al. report postfixation volumes for

the WB, medulla oblongata, CB, mesencephalon, diencephalon, and telencephalon. These volumes were estimated by weighing cuttings from photographs of serial sections. For purposes of determining allometric exponents, we consider these data the equivalent of our weight data. Stephan et al. corrected brain volumes for shrinkage, based on specific gravity information. Our data, on the other hand, were not corrected, because only relative brain volumes/weights are required for the purposes of this study.

The data from Stephan et al., for the medulla oblongata and the mesencephalon, are collapsed into BS data, and their data from the diencephalon and the telencephalon are collapsed into FB data, for the sake of comparing data from our two studies. All brain weight and volume data are collapsed across subjects within each species, giving a mean for the species. Species means are transformed into common logarithms [for a discussion of the appropriateness of this transformation, see Smith, 1980 and Harvey, 1982]. Linear regression analyses (method of least squares) were performed on these transformed data, paired with transformed body weight data. For the purposes of this study, only the slopes of the regression lines are important. Comparisons of groups of three or more slopes are performed using analyses of variance, and comparisons of pairs of slopes are performed using Tukey-Kramer and T' minimum significant differences (MSDs) [see Sokal and Rohlf, 1981]. For comparisons between pairs of regression lines within common groups (and species), the two MSD methods are mathematically identical.

Results

The means and SDs for all rodent measurements are given in table I, and the data for insectivores and primates are found in Stephan et al. [1970]. Common logarithms of mean values for rodent SCA, WB weight, CB weight, BS weight, and FB weight are plotted against the log body weight for each rodent species in figure 2. In figure 3, the mean volumes for the above brain divisions (excluding the SCA) in insectivores and primates are

Table I. Rodent weight data (means $\pm$ SD)

Species	Weight, g					SCA mm ²
	body	WB	BS	CB	FB	
F mouse	31.1 $\pm$ 2.8	0.613 $\pm$ 0.036	0.134 $\pm$ 0.009	0.078 $\pm$ 0.007	0.399 $\pm$ 0.026	4.89 $\pm$ 0.375
M mouse	32.8 $\pm$ 3.5	0.623 $\pm$ 0.027	0.132 $\pm$ 0.007	0.074 $\pm$ 0.006	0.402 $\pm$ 0.021	4.76 $\pm$ 1.08
M gerbil	82.5 $\pm$ 10.5	0.977 $\pm$ 0.067	0.224 $\pm$ 0.018	0.119 $\pm$ 0.010	0.634 $\pm$ 0.085	5.83 $\pm$ 1.08
M hamster	132 $\pm$ 13	1.19 $\pm$ 0.04	0.237 $\pm$ 0.015	0.141 $\pm$ 0.008	0.798 $\pm$ 0.027	6.67 $\pm$ 0.50
F SD rat	423 $\pm$ 63	2.46 $\pm$ 0.13	0.532 $\pm$ 0.037	0.309 $\pm$ 0.027	1.63 $\pm$ 0.10	12.12 $\pm$ 0.80

F = Female; M = male; SD = Sprague-Dawley.

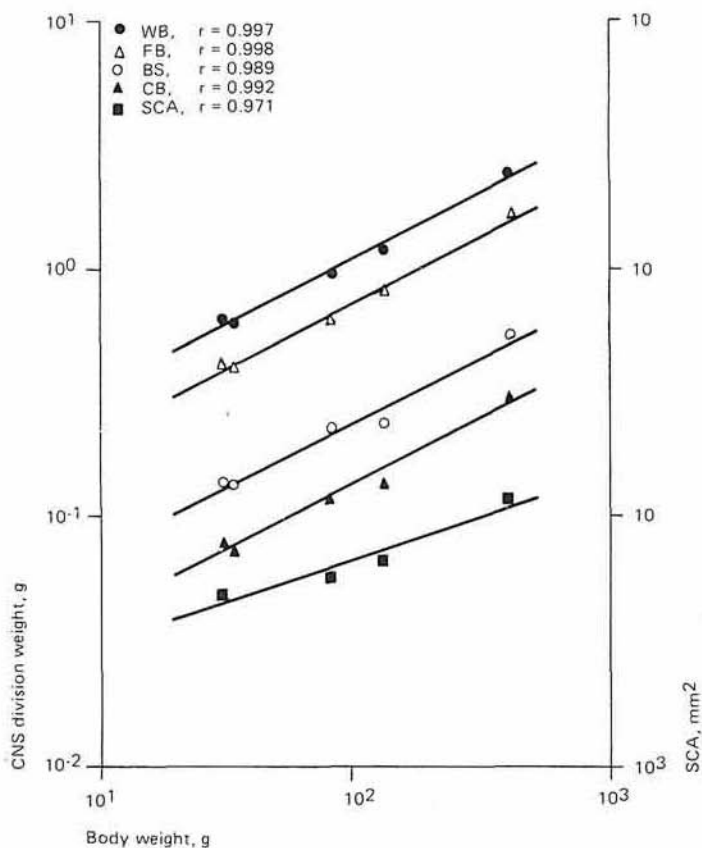


Fig. 2. log/log plots of all rodent brain measures against body weight. Regression lines and correlation coefficients are given.

Table II. The exponents of allometry for the various brain regions for each animal grouping

Brain area	Rodents	Basal insectivores	Progressive insectivores	Prosimians	Simians
WB	0.526 ± 0.073	0.639 ± 0.056	0.664 ± 0.135	0.654 ± 0.101	0.742 ± 0.100
SCA	0.341 ± 0.155	NM	NM	NM	NM
CB	0.528 ± 0.123	0.676 ± 0.078	0.723 ± 0.138	0.692 ± 0.098	0.790 ± 0.094
BS	0.520 ± 0.141	0.593 ± 0.055	0.679 ± 0.150	0.529 ± 0.079	0.511 ± 0.058
Medulla	NM	0.590 ± 0.058	0.678 ± 0.135	0.567 ± 0.074	0.525 ± 0.059
Mesencephalon	NM	0.602 ± 0.059	0.671 ± 0.231	0.479 ± 0.092	0.493 ± 0.064
FB	0.536 ± 0.065	0.644 ± 0.065	0.645 ± 0.153	0.735 ± 0.176	0.751 ± 0.105
Diencephalon	NM	0.626 ± 0.079	0.662 ± 0.197	0.657 ± 0.094	0.612 ± 0.078
Telencephalon	NM	0.646 ± 0.064	0.643 ± 0.150	0.766 ± 0.242	0.761 ± 0.107

All data are analyzed against body weight. For each exponent, the 0.05 confidence limits are given. NM = Measurements were not made in these brain areas.

plotted against log body weight. Regression lines are superimposed over figures 2 and 3, along with correlation coefficients. The exponents (slopes) of these regression lines, along with $p < 0.05$ confidence intervals, are listed in table II.

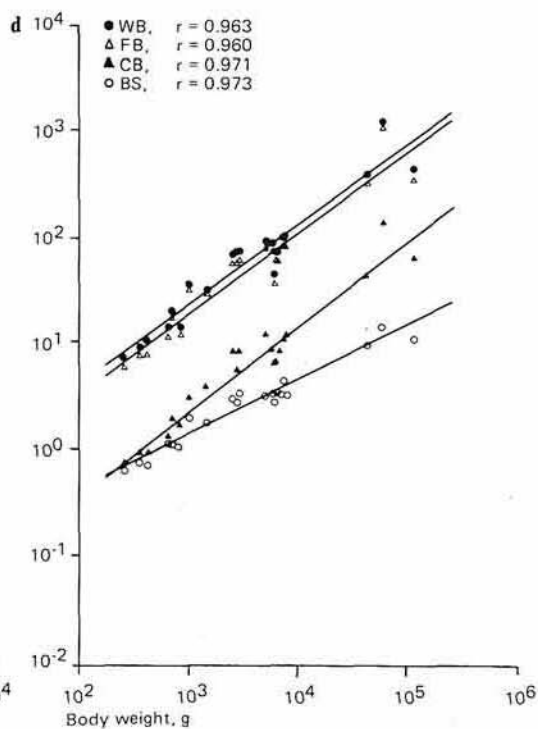
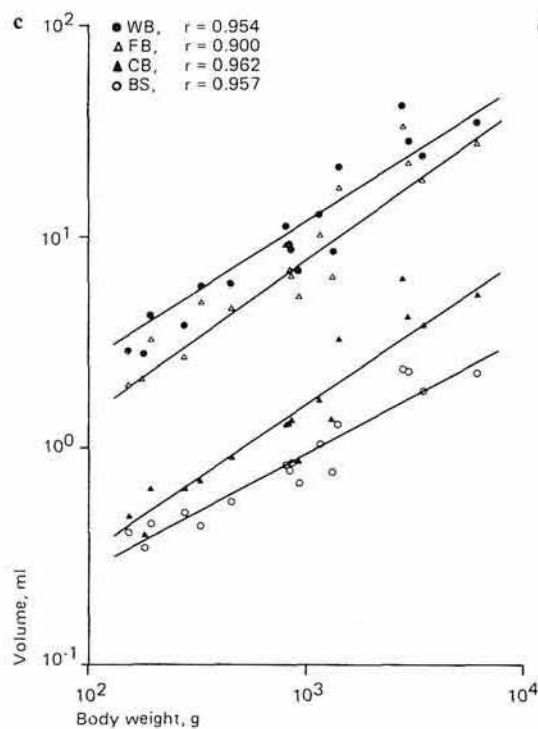
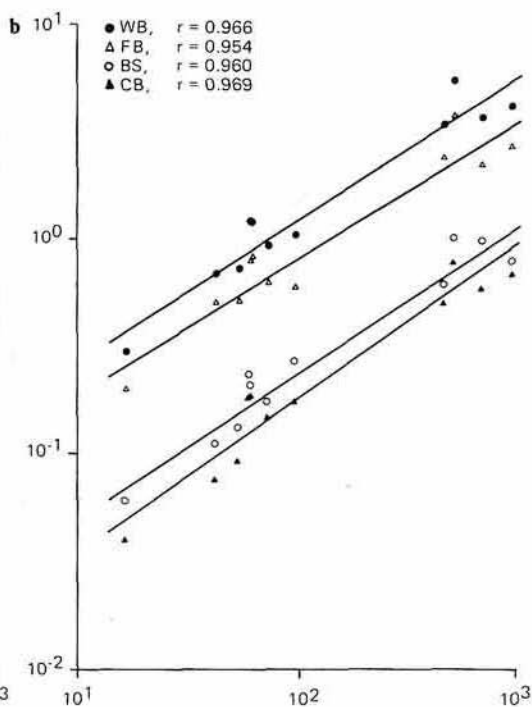
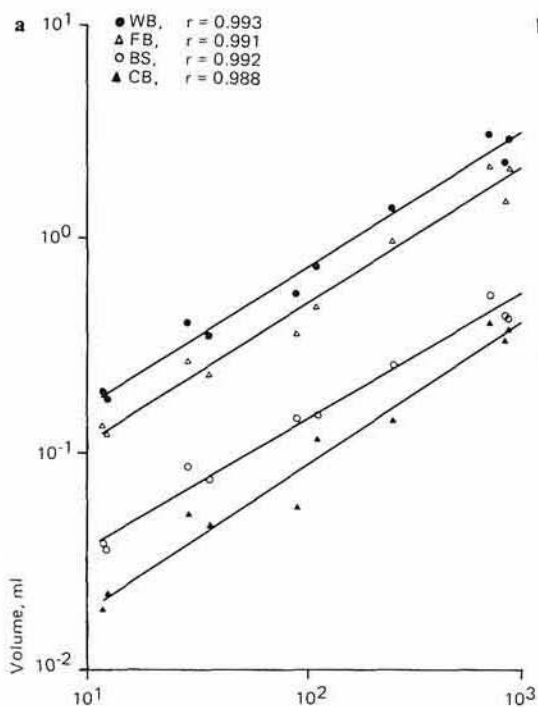
The exponents of variation of BS weight with body weight for the five animal groups are approximately 0.52, with the exception of the progressive insectivores. Their BS exponent, 0.679, is significantly greater than that of simians, 0.511 (the lowest exponent), when a Tukey-Kramer MSD is applied ($MSD = 0.162$ for $p < 0.05$), but not when a T' MSD is applied ($MSD = 0.200$ for $p < 0.05$). No other differences found between any animal groups and within any given measure were significant by any test.

The exponents for the three CNS divisions did not differ significantly within either rodents ($p \geq 0.25$), basal insectivores ($p > 0.10$) or progressive insectivores ($p \geq 0.25$). Exponents did, however, differ between divisions within simians

($p < 0.05$) and prosimians ($p < 0.01$). Among prosimians, the BS scaled by a lower exponent ($a = 0.529 \pm 0.079$) than did the FB ($a = 0.735 \pm 0.176$; $MSD = 0.205$ for $p < 0.05$), and among simians, the BS scaled by a lower exponent ($a = 0.511 \pm 0.058$) than either the FB ($a = 0.751 \pm 0.105$) or CB ($a = 0.790 \pm 0.094$; $MSD = 0.183$ for $p < 0.01$).

Where analyses of variance indicated homogeneity of slope within a set of comparisons, a common slope was determined. Slopes common to the BS, CB, and FB measures were 0.528 in rodents, 0.638 in basal insectivores, and 0.683 in progressive insectivores. And slopes common to all five animal groups are 0.724 for CB and 0.703 for FB.

Those exponents of allometry differing significantly from the value $2/3$ are as follows (all tests 2-tail): rodents WB ($a = 0.526 \pm 0.073$, $t = 6.115$, $p < 0.01$), BS ($a = 0.520 \pm 0.141$, $t = 3.308$, $p < 0.05$), CB ($a = 0.528 \pm 0.123$, $t = 3.584$, $p < 0.05$), FB ($a = 0.536 \pm 0.065$, $t = 6.428$, $p < 0.01$),



and SCA ($a = 0.341 \pm 0.155$, $t = 6.688$, $p < 0.01$); *basal insectivores* BS ($a = 0.593 \pm 0.055$, $t = 3.020$, $p < 0.02$); *prosimians* BS ($a = 0.529 \pm 0.079$, $t = 3.663$, $p < 0.002$); *simians* BS ($a = 0.511 \pm 0.058$, $t = 5.601$, $p < 0.002$) and CB ($a = 0.790 \pm 0.094$, $t = 2.753$, $p < 0.02$).

Discussion

The primary question this study addresses is whether the exponent of allometry for somatic brain regions, a_v , may be treated the same as that for nonsomatic regions, a_c . Clearly, the exponents of allometry differ significantly between different brain regions. Generally, the FB and the CB scale by the highest exponent of body weight, and the BS scales by a lower exponent. The one exception to this trend is the progressive insectivore group, whose BS scales very slightly higher than its FB (but lower than its CB). Regional differences are greatest in simians, less in prosimians, less in insectivores, and practically nonexistent in rodents. Clearly, if these divisions differ by degree of somatic orientation, then their exponents of allometry would be equal *if and only if* $a_c = a_v$. For both prosimians and simians, exponents for the three divisions differ significantly from each other ($p < 0.05$). Because of these differences, we conclude that a_c is not always equal to a_v . Furthermore, the BS, which probably has the

most somatic involvement of the three divisions [see Hofman, 1982; Krompecher and Lipak, 1966], generally varies by a lower exponent than the other two divisions. If a_v were greater than a_c , then the more somatic divisions would vary by a *higher* exponent. Therefore, we tentatively conclude that $a_v < a_c$. And because $a_v > 0$, then $a_c > 0$. This is contrary to the notions of nonsomatic brain scaling [e.g. by Jerison, 1973] on which much of brain allometry research has been predicated. The cumulative weight of the nonsomatic brain regions, E_c , does vary systematically with body weight by a different exponent than does the cumulative weight of the somatic brain regions, E_v . As a result, equation (2) is too simple to represent brain/body relationships accurately. In order to avoid most of the theoretical and descriptive difficulties associated with the differential contributions of E_c and E_v to E (see Introduction), it is necessary to consider these factors independently. We, therefore, propose the following relationships: both E_v and E_c probably vary by simple exponents of P ; thus,

$$E_v = k_v P^{a_v}, \text{ and} \quad (5)$$

$$E_c = k_c P^{a_c}, \quad (6)$$

where k_v is a constant reflecting the magnitude of somatic scaling, and k_c is the new 'index of cephalization'. Substituting equations (5) and (6) into equation (4):

$$E = k_v P^{a_v} + k_c P^{a_c}. \quad (7)$$

It should be noted that we do not consider the above equation to be 'the correct' brain allometry equation. We have merely distinguished between two somewhat independent factors that are involved in the scaling of the whole brain. Other factors

Fig. 3. log/log plots for basal insectivores a, progressive insectivores b, prosimians c, and simians d of WB, BS, CB, and FB weights, against body weight. Regression lines and correlation coefficients are given.

may still exist. For instance, there may be a difference in scaling between sensory and motor systems, or between sensory systems of different modalities (e.g. visual vs. tactile). However, any such distinctions are probably of less consequence than the distinction considered in this study.

Even though equation (7) is a conceptual improvement over equation (2), it is not obvious what the values should be for k_v , a_v , and a_c . Although the BS is believed to represent mostly somatic involvement, it is not necessarily valid to consider its exponent of allometry, a_{BS} , equal to a_v [Jernison, 1970]. Whether this equivalence may be drawn depends partially on the consistency of the value of a_{BS} between different groups of animals. Probably animals of a similar body plan share a fundamental biological relationship between E_v and P . For this reason, if different animal groups with similar body plans have different values for a_{BS} , then it is not likely that a_{BS} may be substituted for a_v . An examination of the data in table II indicates that a_{BS} is not consistent across animal groups (although it does not vary significantly). Although a_{BS} is high for progressive (0.593) and basal (0.679) insectivores in this study, Wirz [1950] found an exponent of 0.48 for the insectivore a_{BS} . Why these discrepancies exist is not clear, although they might result from the exclusion of part of the mid-brain tissue from the BS measurement in the Wirz study. But because these discrepancies exist, a_{BS} alone may not be used as a reliable estimator of a_v .

A value for a_v , however, may be derived from the rodent data as follows. Within the rodent species studied, the exponents for the different brain weight measures (i.e. excluding a_{SCA}) are all ap-

proximately equal, differing by no more than 0.016. Thus, throughout rodents, major brain divisions retain certain fixed proportions with respect to the WB. Based on the assumption that the brain divisions used in this study differ from each other in their somatic and nonsomatic content and based on the finding that the exponents for rodent brain divisions are roughly equal, it is possible to conclude that for rodents, a_v is roughly equal to a_c . (If this were not true, then the brain divisions would not vary by a common exponent.) Thus, for rodents, nonsomatic areas vary with body weight just as though they were somatically involved. The exact cause for this is irrelevant for the purposes of this study. What is important is that, by coincidence, each division of the rodent brain, despite its proportion of somatic and nonsomatic tissue, varies with body weight as though it were entirely somatically involved. Thus, a_v is equal roughly to the exponent for each separate brain region, as well as that for the whole brain. Because there is minute variation in exponents, however, it is likely that the best guess for a_v is the exponent for the most somatic brain region (i.e. the BS). Here, the effect of a_c is minimized. Therefore, we estimate a_v to be 0.520. This value is very similar to those obtained for a_{BS} for prosimians (0.529) and simians (0.511). Also, it is very similar to measures by Portmann [1947] for birds (0.52) and by Wirz [1950] for insectivores (0.48), and it does not differ significantly from BS exponents for either basal or progressive insectivores. Unlike the exponent of $2/3$, 0.520 has no obvious and/or simple meaning; rather, it is useful for descriptive purposes only. Although the value of a_v found for rodents might

safely be generalized to the above animals, caution should be taken in applying this value to animals with body plans dissimilar to those of rodents (especially with dissimilar sensory and motor systems).

Because only four species were used to determine rodent allometry exponents, the results of this study may not be representative of the entire rodent order (and strictly should not be considered as such). As a result, there may actually be variation in brain scaling between rodent brain divisions. Only because our sample yielded equal exponents were we able to deduce a value for a_v which might be roughly applicable to other mammals. But because only four species were used, the confidence intervals for the exponents are wide; consequently, our estimate of a_v may not be accurate. If larger rodent samples reveal discordant exponents (which would not facilitate a valid estimation of a_v), then perhaps a large sample of mammals, both rodent and nonrodent, should be selected for analysis on the basis of similarity in both brain proportioning and body plan. Theoretically, *any* mammalian sample that meets these criteria should yield the same allometry exponents for each brain division and, hence, for a_v . Of course, the sampling of wide taxonomic ranges in itself increases the ability to generalize to unsampled species.

The secondary goal of this study is to reevaluate whether body surface area (which varies by the $2/3$ power of body weight) is a valid determinant of total somatic innervation and, eventually, brain weight. According to our measurements, the exponent of allometry for the rodent SCA is 0.341, and the $p < 0.05$ confidence limits are 0.186 and 0.496. We interpret

this to mean that within the rodent species examined, the total number of spinal afferents and efferents varies by the 0.341 power of body weight. An estimate made by Hofman [1982] for this measure in mammals, 0.403, falls within the confidence interval. Clearly, $2/3$ falls well outside of this interval ($p < 0.01$), as do numerous weight and volume measurements listed previously. Thus, we cannot support the hypothesis that the total number of somatic afferents and efferents varies by a direct proportion of body surface area. As previously mentioned, Jerison [1979] indicates that rodent brains in particular should exhibit $2/3$ power scaling, because the sizes of their cortical maps tend to vary with brain/body size, while overall cortical organization appears constant. Although we had no measure of cortical organization, we did find that the rodents examined exhibited a constant gross brain proportioning (indicated by similar exponents of allometry for the major CNS divisions), but neither WB nor the SCA scale by $2/3$.

The degree to which the SCA reflects the total number of somatic afferents and efferents is not entirely clear. Specifically, two problems may exist for this measure. First, the SCA measure includes the cross-sectional areas of both the spinal white matter (representing fiber tracts) and the gray matter (representing nuclear groups). Strictly speaking, only the cross-sectional area of the spinal white matter should have been used. We made our measurements, however, assuming that the number of cell bodies in the gray matter must be roughly proportional to the total somatic innervation and minimized the contributions of gray matter by making our

measurements at upper cervical levels. A finer analysis may show spinal white matter to vary by the $2/3$ power of body weight, but we feel this to be unlikely.

The second difficulty with the SCA measure is that the density of spinal fibers may correlate negatively with their length and, hence, with the size of the animal. Such a relationship would result in decreased fiber density with increased body size. As a result, the exponent of allometry for the number of spinal fibers would not accurately be reflected by the SCA exponent. Comparing the cross-sectional areas and fiber counts of the pyramidal tracts (rostral medulla) in 6 ungulates (deer, mule, cow, goat, sheep, and hog), reported by Lassek [1942], with the average body weights for these species [as listed by Spector, 1956] reveals tentatively that fiber density does not seem to vary systematically with body weight.

But if fiber diameter is positively correlated with body size, then total cross-sectional area should also be positively correlated, after fiber count is parcelled out. As a result, the SCAs for larger species would tend to be larger than the fiber counts would indicate. Consequently, the exponent of allometry for fiber counts would be lower than that for the SCA. This would mean that for the rodents measured, the exponent for the somatic fiber count, as a function of body weight, has a value no greater than 0.496 (the upper confidence interval limit for the SCA) but possibly lower than 0.186 (the lower limit). Consequently, we reject the hypothesis that body surface area is a simple (directly proportional) determinant of total somatic innervation. This is not to say that surface area does not relate to brain size in

any manner, but probably too many factors determine total innervation needs for surface area to be a simple, useful, or theoretically valid single predictor of brain size. Thus, if the brains of any set of animals scale to the $2/3$ power of body weight, we feel that such scaling is entirely coincidentally (rather than causally) determined.

In summary, this study has defended the following: (1) body surface area is not a direct and simple determinant of somatic information that must be processed by the brain, theoretically invalidating the surface area account of brain allometry; (2) the size of the nonsomatic aspect of a brain is not independent of body weight; (3) somatic brain regions scale by a different exponent of body weight than do nonsomatic regions, necessitating the abandonment of equation (2), in favor of equation (7), and (4) the exponent of allometry for somatic brain regions is approximately 0.520.

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